fastapi

ℹ *This wasn't fully covered in your uploaded materials, so I also used web sources.*

FastAPI is a **modern Python web framework** for building **APIs** (Application Programming Interfaces) and backend services. Think of a web framework as a toolkit that handles the low-level details of receiving HTTP requests, routing them to your code, and sending back responses, so you can focus on writing your application logic.

Core ideas from the sources:

- **Python 3.8+ and type hints** – FastAPI is built around standard Python **type hints** (the optional annotations like `def func(x: int) -> str:`). These hints let the framework automatically validate, serialize, and deserialize data [1].
- **Pydantic** – A library that uses those type hints to define data schemas, perform validation, and convert data between Python objects and formats like JSON [1].
- **Starlette** – A lightweight **ASGI** (Asynchronous Server Gateway Interface) toolkit that gives FastAPI its high-performance, asynchronous request handling [2].
- **Automatic interactive documentation** – FastAPI generates **Swagger UI** and **ReDoc** pages automatically from your code, so you can explore and test your API in a browser without writing extra documentation [3].

Key features (summarized from the official site and tutorials):

- **High performance** – On par with NodeJS and Go because of Starlette and Pydantic [2].
- **Fast development** – Claims to increase feature development speed by 200-300 % [2].
- **Fewer bugs** – Type-driven validation catches many errors early, reducing bugs by roughly 40 % [2].
- **Async support** – Native `async/await` syntax lets you handle many concurrent connections efficiently [3].
- **Beginner-friendly syntax** – Simple, intuitive decorators (e.g., `@app.get("/items")`) make it easy to define routes [4].
- **Production-ready** – Used for REST APIs, microservices, and backends for web/mobile apps [4].

In a nutshell: FastAPI lets you write clean, type-annotated Python functions, and it takes care of request parsing, validation, serialization, and even generates live API docs for you. It's designed to be both fast to run and fast to write.

Sources

1. https://en.wikipedia.org/wiki/FastAPI
2. https://fastapi.tiangolo.com/
3. https://www.geeksforgeeks.org/python/fastapi-tutorial/
4. https://www.geeksforgeeks.org/python/introduction-to-fastapi/